MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: DESIGN CALCULATION 1800 SECURITY BASE CAMP FIRE FIGHTING PUMP HOUSE DESIGN OF BUILDING STRUCTURE

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
Typ.	:	Typical
HCS	:	Hollow Core Slab
Fc'	:	Concrete Compressive Strength
Fy	:	Yield Strength of Reinforcing Steel
ϕ	:	Strength Reduction Factor
SBC	:	Saudi Building Code
SASO	:	Saudi Standards, Metrology and Quality Organization
ACI	:	American Concrete Institute
ASCE	:	American Society of Civil Engineers
PCI	:	Precast/Prestressed Concrete Institute

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1. INTENT

This Document is intended to produce the technical needs in Engineering including particular reference to design justification & Engineering validation works. The work described below is related to Design of Fire Pump House in Security Base Camp Area.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The Scope of this document is limited to Complete Analysis & Structural Design Verification of Fire Pump House Building only.

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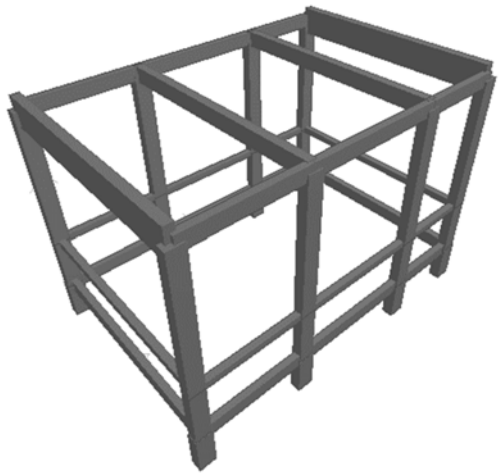
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4. DESIGN METHODOLOGY & ASSUMPTIONS

This Building Is a Single Story Frame Structure with Foundation Being Cast in Site. The Building Shall Have Insulated Block Walls and Cast in Site Framing Beams to support the HCS (Hollow Core Slab)

The Superstructure for this Accommodation Building was modeled using STAAD software. The Analysis of Frames (Cast-In-Situ Columns & Beams) has been performed meeting design requirements & also keeping in view the Architectural requirements. The loads were applied as per SBC/ASCE/ACI standards & analysis was done to generate the Column Reactions at the Base which are assumed to be fixed. The Foundation design has been done by importing Column Base Reactions into Foundation Design Software.



STAAD 3D MODEL

4.1. Codes

[1] SBC-301-2007-Structural Loading & Forces.

[2] SBC-303-2007-Structural Soil & Foundations.

[3] SBC-304-2007-Structural Concrete Structures.

[4] SASO SSA 2 – Steel Bars for the Reinforcement of Concrete.

[5] SASO SSA 224 – Steel Fabric for Reinforcement of Concrete

[6] ASCE-7-2005 Minimum Design Loads for Buildings and other Structures.

[7] ACI 318-08 Building Code Requirements.

[8] PCI –MNL-120-04 –Design Handbook, 6th edition.

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4.2. Geotechnical investigations

Foundations have been designed based on recommendations of the Soil Investigation Report, Doc # MD-216-3000-EG-CV-RPT-1008.

4.3. Material Properties

Compressive Strength of Concrete at 28th days

fc' = 35 MPa (5.00 Ksi) for beams, columns, suspended slabs.

fc' = 35 MPa (5.00 Ksi) for foundations, grade beams, columns.

Yield Strength of Steel

fy = 414 MPa (60 Ksi) for all reinforcing steel

5. DESIGN LOADS

5.1. Dead Load

5.1.1. Roof Dead Loads

- Provision for Electrical Lighting = 0.05 kN/m²

Ceiling

Suspended steel channel system = 0.10 kN/m²

Acoustical Fiber Tile (16mm) = 0.05 kN/m²

Sub-total = 0.2 kN/m²

- Hollow Core Slab (PCI 6th Edition)

(HCS-250 Topped) = **3.43 KN/m2**

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Prestressed Hollowcore Floor										
Overall Structural Depth (mm)	Self Wt- KN/M2	No. of Cores in Unit	Spans indicated below allow for Characteristic Service Load (Live Load) Plus Self Weight Plus 1.5Kn/m² for Finishes Characteristic Service Loads Kn/m²							
			0.75	1.50	2.00	2.50	3.00	4.00	5.00	10.00
			Clear Span in metres							
100	2.38	0	5.39	5.19	5.08	4.91	4.72	4.40	4.14	3.30
150	2.52	11	7.40	7.40	7.40	7.40	7.28	6.80	6.40	5.12
150 (Dense	2.94	11	7.40	7.40	7.40	7.35	7.08	6.64	6.27	5.06
200	3.10	11	9.40	9.40	9.40	9.40	9.26	8.69	8.21	6.65
250	3.43	6	10.80	10.80	10.80	10.80	10.80	10.80	10.56	8.60
300	3.88	6	13.50	13.50	13.50	13.50	13.18	12.44	11.81	9.70
350	4.23	6	15.00	15.00	15.00	15.00	14.89	14.08	13.39	11.06
400	4.57	6	16.00	16.00	16.00	16.00	15.97	15.13	14.42	11.98
450	4.92	6	18.00	18.00	18.00	17.70	17.19	16.33	15.58	13.02
500	5.27	6	18.00	18.00	18.00	18.00	18.00	17.13	16.38	13.76

- Built-in Water Proofing System

100mm thk Polystyrene Foam

0.01 kN/m² per 25.4mm thickness x (100/25.4mm) = 0.039 kN/m²

Water proofing membrane

Bituminous, smooth surface = 0.07 kN/m²

90 mm thick average cement foam concrete screed Lightweight

Concrete 0.38 kN/m² per 25mm thickness x (90/25mm) = 1.368 kN/m²

75 mm Concrete screed above HCS = 21 x 0.075 =1.6 kN/m²

Sub-total = **3.1 kN/m²**

Total Load = **6.53 kN/m²**

UDL on External Beams along grid 1& 2 = 8.20 x 6.53/2 = **26.77 Kn/m**

Moment on all the External Beams due to Cantilever Portion of slab and Screed

= (0.15x24+ (0.05+0.2)/2x20)*0.6*0.6/2 = **1 knm/m**

5.1.2. Ground Floor Walls

- 250 thk. Outer CMU

16 x 0.25 x 6.5m clear height = **26.0 kN/m**

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5.2. Live Load

5.2.1. Roof Live Loads

Live loads were taken from SEI/ASCE 7-05, Table 4-1.

Ordinary Flat = **0.50 kN/m²**

UDL on External Beams along Grid Line-2 = 0.50 x 8.20/2 =**2.05 Kn/m**

Moment on all the External Beams due to Live Load on

Cantilever Portion of slab = 1*0.6*0.6/2 = **0.18 kN/m /m**

5.3. Wind Load

Wind loads were automatically generated in STAAD as per ASCE 7-05. The parameters used for the wind load generation are as follows (Refer STRUCTURAL DESIGN CRITERIA NON-PROCESS BUILDINGS Doc No MD-216-0000-EG-ST-CRT-0001)

All structures and foundations (except for buried structures) designed and shall be constructed to resist the wind effects determined in accordance with Section-7.2, SBC-301-Analytical Procedure.

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5.4. Wind Load (Analytical Verification)

WIND LOADING (SBC-301)

Using the directional design method

Building data

Type of roof;	Flat
Length of building;	b = 12000 mm
Width of building;	d = 8000mm
Height to eaves;	H = 7250 mm
Mean height;	h = 7250 mm

General wind load requirements

Basic wind speed;	V = **49** m/s (or) 177 kmph (Refer Clause-4.2.6, Doc- MD-216-1000-EG-ST- CRT-1001)
Risk category;	II(As per SBC-301, Table 1.6-1)
Velocity pressure exponent coeff;	K_d = 0.85(As per SBC-301, Table 6.4-1)
Exposure category ;	D(As per SBC-301, Table 6.4.2.3)
Enclosure classification ;	Enclosed buildings(As per SBC-301, Section-6.2)
Internal pressure coef +ve;	GC_{pi_p} = 0.18(As per SBC-301, Figure-7.2-1)
Internal pressure coef –ve;	GC_{pi_n} = -0.18(As per SBC-301, Figure-7.2-1)
Gust effect factor;	G_f = 0.85(As per SBC-301, Section-7.2.7.1)
Importance Factor;	1(As per SBC-301, Section-9.5)

Topography

Topography factor not significant;	K_{zt} = 1.0(As per SBC-301, Section-6.4.3.2)
Velocity pressure equation;	$q = 0.0473x10^{-3}Kn/m^2 \times K_z \times K_{zt} \times K_d \times V^2 \times I$ V in Km/hr(As per SBC-301, Section-7.2.8)

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Velocity pressures table

z (mm)	K_z (SBC-301,Table 7.2-2)	q_z (kN/m^2)
4572	0.85	1.06
7250	0.93	1.06

Pressures and forces

Net pressure; $p = q \times G_f \times C_{pe} - q_i \times GC_{pi}$;

Net force; $F_w = p \times A_{ref}$;

Walls load case 1 - Wind 0, GC_{pi} 0.18, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A1	4.57	0.80	1.06	0.51	54.86	28.18
A2	4.57	0.80	1.06	0.51	0.00	0.00
A3	7.25	0.80	1.16	0.58	32.15	18.71
B	7.25	-0.50	1.16	-0.70	87.00	-61.26
C	7.25	-0.70	1.16	-0.90	58.00	-52.31
D	7.25	-0.70	1.16	-0.90	58.00	-52.31

Walls load case 2 - Wind 0, GC_{pi} -0.18, + c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A_1	4.57	0.80	1.06	0.93	54.86	51.17
A_2	4.57	0.80	1.06	0.93	0.00	0.00
A_3	7.25	0.80	1.16	1.00	32.15	32.18
B	7.25	-0.50	1.16	-0.29	87.00	-24.81
C	7.25	-0.70	1.16	-0.48	58.00	-28.01

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Walls load case 3 - Wind 90, GC_{pi} 0.18, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A_1	4.57	0.80	1.06	0.51	36.58	18.79
A_2	4.57	0.80	1.06	0.51	0.00	0.00
A_3	7.25	0.80	1.16	0.58	21.43	12.47
B	7.25	-0.40	1.16	-0.61	58.00	-35.10
C	7.25	-0.70	1.16	-0.90	87.00	-78.47

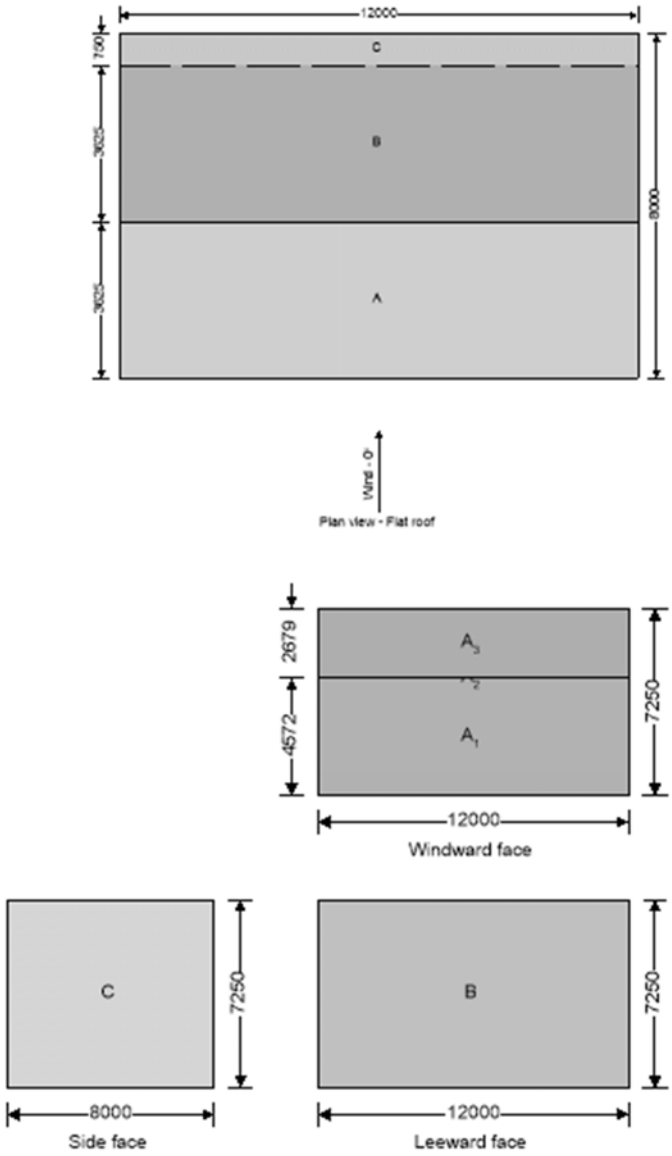
Walls load case 4 - Wind 90, GC_{pi} -0.18, + c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A_1	4.57	0.80	1.06	0.93	36.58	34.11
A_2	4.57	0.80	1.06	0.93	0.00	0.00
A_3	7.25	0.80	1.16	1.00	21.43	21.45
B	7.25	-0.40	1.16	-0.19	58.00	-10.80
C	7.25	-0.70	1.16	-0.48	87.00	-42.02

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So from the above analysis, the Wind Loads generated in STAAD are in accordance with SBC-301-2007-Structural Loading & Forces.

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5.5. Seismic Load

Earthquake loads were automatically generated in STAAD as per IBC 2003. The basic parameters affecting the generation of seismic loads is shown below:

Importance Factor, I = 1.0(As per SBC-301, Section-9.5)

Design Spectral Response Acceleration (Refer Clause-4.2.7b, MD-216-1000-EG-ST- CRT-1001)

Ss = 0.1g (Spectral Response Acceleration 0.2 second)

S1 = 0.02g (Spectral Response Acceleration 1 second)

S_{D_1} = 2/3*Fv*S1 = 2/3 * 2.4 * 0.02 = 0.032 (As per SBC-301, Section-9.4.4)

S_{DS} = 2/3*Fa*S1 = 2/3 * 1.60 * 0.10 = 0.10 (As per SBC-301, Section-9.4.4)

Where S_{D_1} Damped Design Spectral Response Acceleration@1 second

S_{DS} Damped Design Spectral Response Acceleration@short period

Site Class = D

Site Coefficient, Fa = 1.60 (As per SBC-301, Table 9.4.3a)

Site Coefficient, Fv = 2.40 (As per SBC-301, Table 9.4.3b)

Building period Coefficient, Ct=0.04 (As per SBC-301, Table 10.9.3.2)

For Ordinary reinforced concrete moment frame,

Response Modification Factor, Rx = 2.5 (As per SBC-301, Table 10.9.3.2)

Response Modification Factor, Rz = 2.5 (As per SBC-301, Table 10.2)

System Over Strength Factor, Omega = 3 (As per SBC-301, Table 10.2)

Deflection Amplification factor, Cd = 2.5 (As per SBC-301, Table 10.2)

5.6. Thermal/Temperature Loads (Self-restraining and Thermal Expansion and Contraction Loads)

An ambient temperature ranges from a maximum of 50° C to a low of 0° C. In direct exposure to sunlight, the temperature shall be assumed to rise a minimum of 15° C above ambient. As a minimum, the temperature gradient for thermal loads shall be +25° C and -15° C with the most severe condition governing the design.

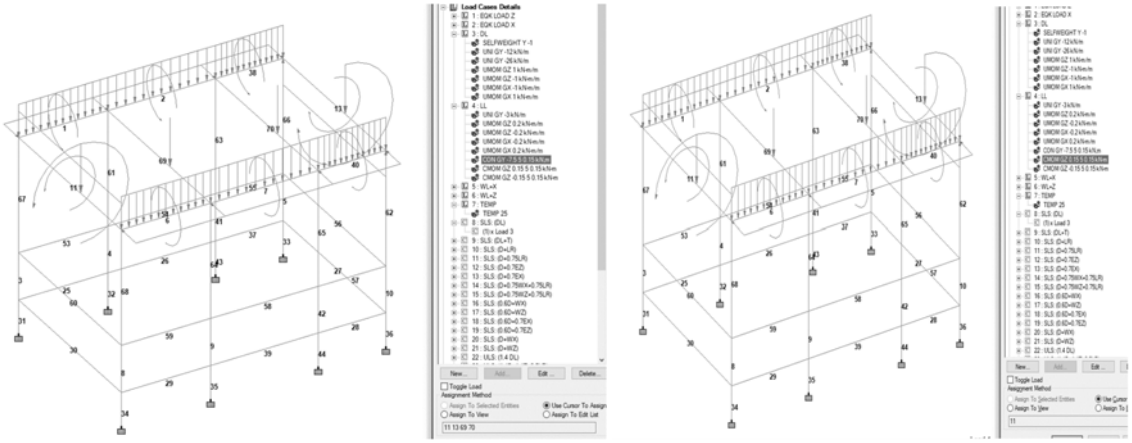
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5.7. Monorail Load

Monorail(1.5 Tons Capacity)working load has been distributed equally on the Supporting concrete beams as shown in the figure below. Also the effect of Longitudional thrust has been considered & moment has been applied with the assumption the Hoist is near towards to the Ends(Beams No-11 & 13)



Total Working Load of the Monorail assumed in the design is 3 Tons.

So Point Load on Each Beam = 30 Kn/4 = **7.5 Kn**

The Beam-B1 provided is well deep enough to undertake all the moments developed due to impact of the Monorails.

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6. DESIGN COMBINATIONS

Service Loads (Allowable Stress Design) (SBC-301, Section 2.4.1)

D

D + T

D +Lr

D +0.75Lr

D + (W or 0.7E)

D + 0.75W + 0.75Lr

0.6D + W

0.6D + 0.7E

Factored loads (Strength Design) (SBC-301, Section 2.3.2)

1.4D

1.4D+1.4T

1.2(D + T) + 0.5Lr

1.2D + 1.6Lr + 0.8W

1.2D + 1.6W + 0.5Lr

1.2D + 1.0E

0.9D + 1.6W

0.9D + 1.0E

Where:

D = Dead Load

Lr = Live Load

W = Wind Load

E = Seismic Load

T = Temperature Load

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7. CRACKED SECTIONS

In lieu of a more accurate analysis of the effects of cracking of concrete, provisions of SBC-304-Section 10.11 was used.

For Beams:	I = 0.35Ig
For Columns:	I = 0.70Ig
For Walls:	
Un-cracked	I = 0.70Ig
Cracked	I = 0.35Ig
For Slabs	I = 0.25Ig

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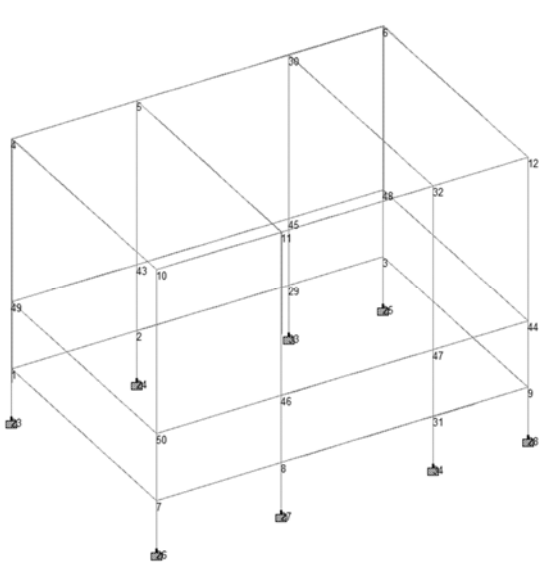
8. DEFLECTIONS/DRIFT CHECK

8.1. Maximum Allowable Deflection

L/240: for member load with live load plus dead load

L/600: for members supporting masonry under total loads

Where, L = span of member between support



FIRE PUMP HOUSE - Node Displacement

Node	L/C	X-Trans mm	Y-Trans mm	Z-Trans mm	Absolute mm	X-Rotan rad	Y-Rotan rad	Z-Rotan rad
32	17	0.304	-0.249	34.317	34.319	0.001	0.001	0.000
32	21	0.370	-0.401	34.316	34.320	0.001	0.001	0.000
5	15	0.025	-0.426	28.876	28.879	0.002	-0.001	-0.000
11	15	0.283	-0.464	28.866	28.871	0.001	-0.001	-0.000
4	21	-0.022	-0.268	27.662	27.663	0.001	-0.001	-0.001
4	17	-0.100	-0.147	27.660	27.660	0.001	-0.001	-0.000
10	17	0.249	-0.215	27.652	27.654	0.000	-0.001	-0.000
10	21	0.327	-0.336	27.650	27.654	0.000	-0.001	-0.001
6	21	0.010	-0.229	26.398	26.399	0.001	0.001	0.000
6	17	-0.054	-0.123	26.396	26.397	0.001	0.001	0.000
12	17	0.325	-0.192	26.389	26.392	0.000	0.001	0.000
12	21	0.389	-0.298	26.387	26.391	0.000	0.001	0.000
30	15	0.024	-0.389	25.638	25.641	0.002	0.001	0.000
32	15	0.294	-0.423	25.628	25.633	0.000	0.001	0.000
4	15	0.020	-0.290	20.678	20.680	0.001	-0.001	-0.001
10	15	0.271	-0.343	20.666	20.670	-0.000	-0.001	-0.001
6	15	0.033	-0.240	19.724	19.725	0.001	0.001	0.000
12	15	0.307	-0.302	19.711	19.716	-0.000	0.001	0.000
5	12	0.165	-0.414	5.432	5.450	0.001	-0.000	-0.000
5	19	0.091	-0.247	5.431	5.437	0.001	-0.000	-0.000
11	19	0.129	-0.255	5.428	5.436	-0.000	-0.000	-0.000
11	12	0.202	-0.422	5.427	5.447	-0.000	-0.000	-0.000
30	12	0.144	-0.377	4.970	4.986	0.001	0.000	0.000
30	19	0.079	-0.225	4.969	4.974	0.001	0.000	0.000
32	19	0.118	-0.232	4.966	4.973	-0.000	0.000	0.000
32	12	0.183	-0.384	4.965	4.983	-0.000	0.000	0.000
4	12	0.177	-0.296	4.237	4.252	0.000	-0.000	-0.000
4	19	0.099	-0.176	4.235	4.240	0.000	-0.000	-0.000
10	19	0.135	-0.187	4.229	4.235	-0.000	-0.000	-0.000
					4.125	0.000	0.000	0.000	0.000
	3 DL				0.000	0.000	0.000	0.000	0.000
					1.031	0.000	-0.291	-0.001	0.291
					2.063	0.000	-0.426	-0.000	0.426
					3.094	0.000	-0.203	0.001	0.203
					4.125	0.000	0.000	0.000	0.000
	4 LL				0.000	0.000	0.000	0.000	0.000
					1.031	-0.000	-0.028	0.001	0.028
					2.063	-0.000	-0.043	0.000	0.043

Maximum attained Drift as per Analysis from above figure = 34 mm

Allowable Drift as per Table-10.12, SBC-301-2007 is H/100 = 7000/100 = 70mm

Hence the Structure is safe considering drift check…

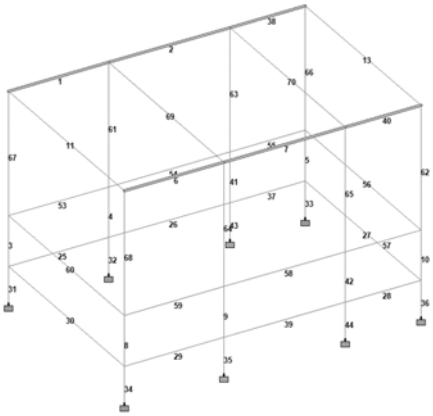
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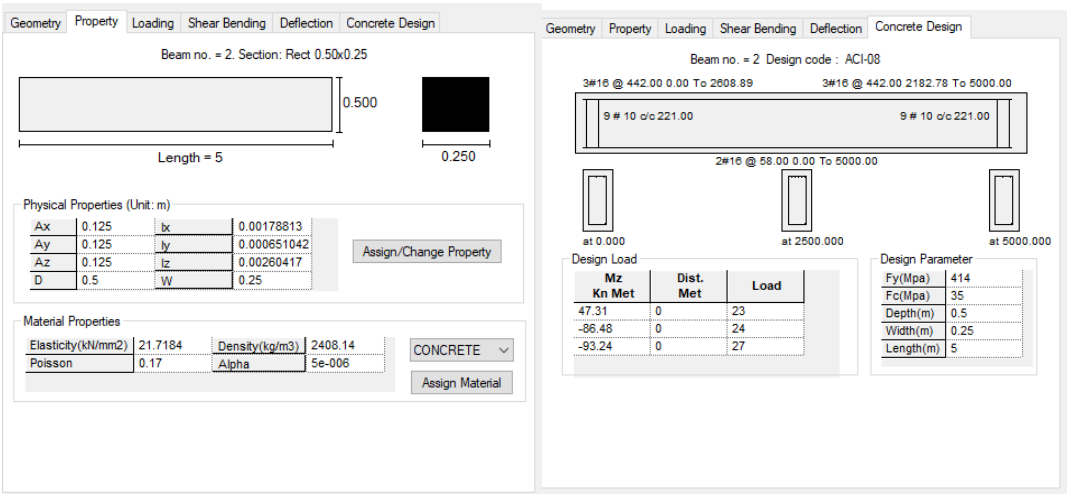
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9. CONCRETE ROOF BEAM DESIGN

9.1. Design of Ledger Beam(LB1)





Design of Ledger Beam (LB1)

Maximum Moment required from Analysis = **93 Knm (Hogging)**

Therefore Ast.req = $$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

Where b=250mm, Fy=414 Mpa, d=500-40-10-20/2=440mm, Fc'=35 Mpa & ϕ =0.9

Ast1.req = **587 mm²**

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Ast.min=
(As per clause-10.5-ACI-318)

$$A_{s,\,min} = \frac{0.25\sqrt{f'_c}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **471mm²**

Maximum Moment required from Analysis = **47 Knm (Sagging)**

Ast2.req = **300 mm²**

However Provide 3-20 dia -Full Straight Bars Top & 3-20 dia-Full Straight Bottom.

Ast1.pro**>>>>>>(Ast.min, Ast1.req)**

Ast2.pro**>>>>>>(Ast.min, Ast2.req)**

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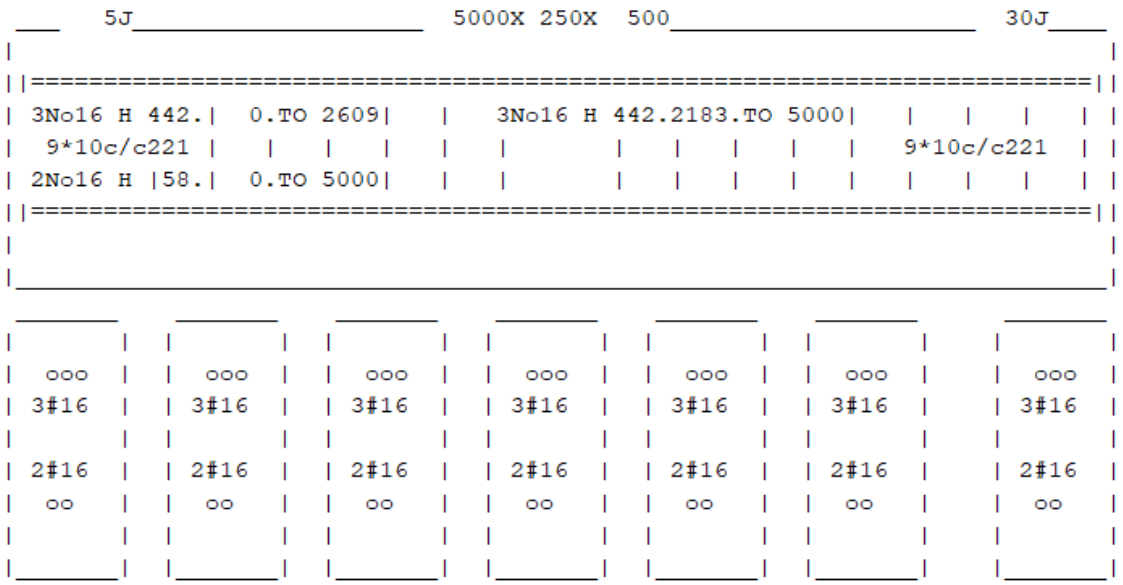
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9.1.1. Staad Output For LB1

```
            B E A M   N O.     2 D E S I G N   R E S U L T S - SHEAR


   AT START SUPPORT - Vu=   52.54 KNS  Vc=  132.11 KNS  Vs=    0.00 KNS
   Tu=    1.94 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD    10
   NO STIRRUPS ARE REQUIRED FOR TORSION.
   REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
   PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM  C/C FOR 1620. MM

   AT END   SUPPORT - Vu=   54.13 KNS  Vc=  120.48 KNS  Vs=    0.00 KNS
   Tu=    1.75 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD    14
   NO STIRRUPS ARE REQUIRED FOR TORSION.
   REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
   PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM  C/C FOR 1620. MM
```

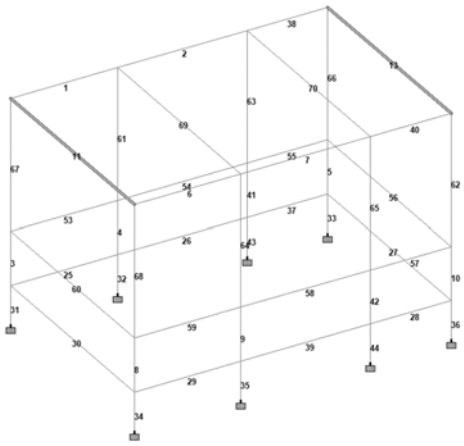


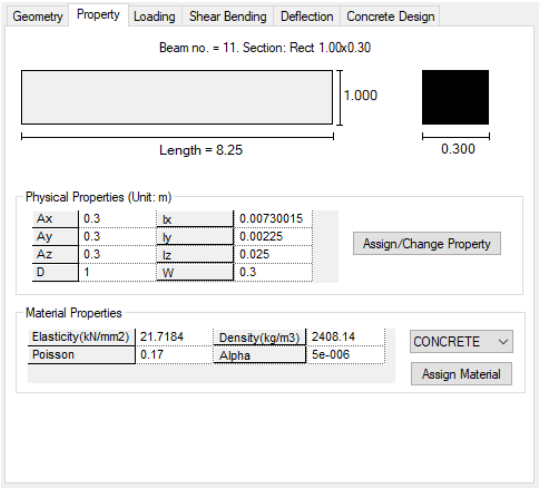
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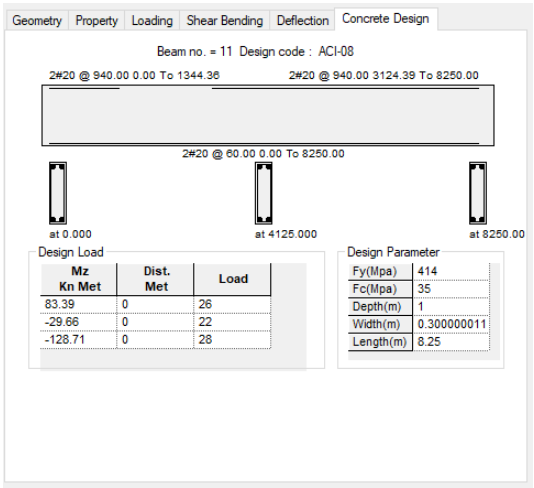
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9.2. Design of Ledger Beam(B1)







Design of Ledger Beam (B1)

Maximum Moment required from Analysis = **128 Knm (Hogging)**

Therefore Ast.req =

$$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

Where b=300mm, Fy=414 Mpa, d=1000-40-10-20/2=940mm, Fc'=35 Mpa & ϕ =0.9

Ast1.req = **365 mm^2**

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Ast.min=
(As per clause-10.5-ACI-318)

$$A_{s,\,min} = \frac{0.25\sqrt{f'_c}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **1000mm²**

Maximum Moment required from Analysis = **83 Knm (Sagging)**

Hence Provide 3-20 dia-Full Straight Bars & 2-16 dia-Extra Bars at Top & 3-20 dia-Full Straight Bars & 2-16 dia-Extra Bars at Bottom considering minimum reinforcement requirements.

Ast1.pro**>>>>>>(Ast.min, Ast1.req)**

Ast2.pro**>>>>>>(Ast.min, Ast2.req)**

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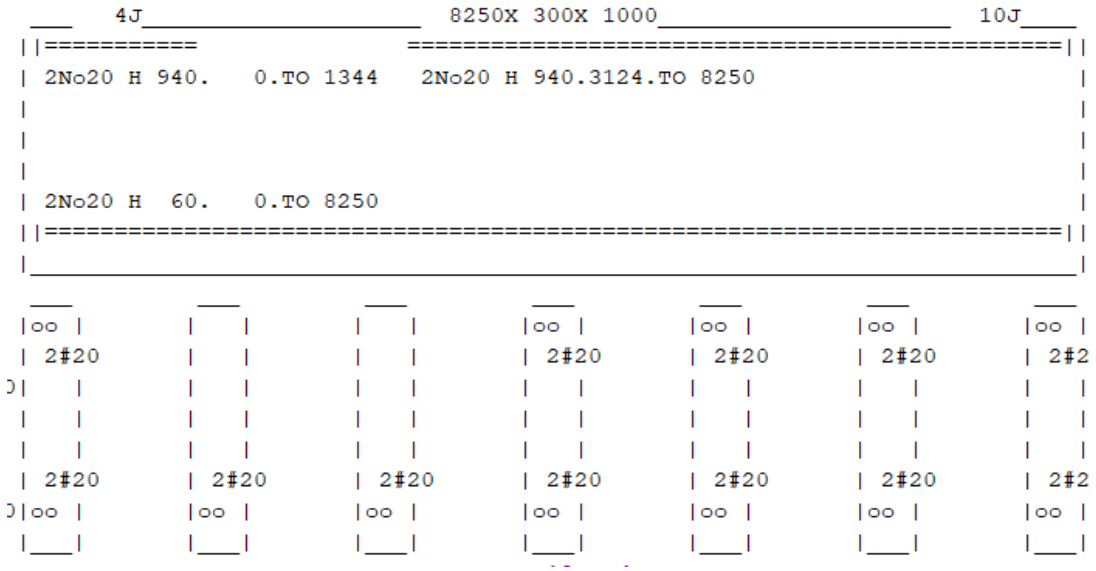
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9.2.1. Staad Output For B1

```
                B E A M   N O.    11 D E S I G N   R E S U L T S - SHEAR

   AT START SUPPORT - Vu=    20.51 KNS   Vc=  281.35 KNS   Vs=     0.00 KNS
   Tu=     2.89 KN-MET   Tc=    12.8 KN-MET   Ts=     0.0 KN-MET   LOAD     3
                       STIRRUPS ARE NOT REQUIRED.
   AT END   SUPPORT - Vu=    20.51 KNS   Vc=  281.35 KNS   Vs=     0.00 KNS
   Tu=     2.89 KN-MET   Tc=    12.8 KN-MET   Ts=     0.0 KN-MET   LOAD     3
                       STIRRUPS ARE NOT REQUIRED.
```

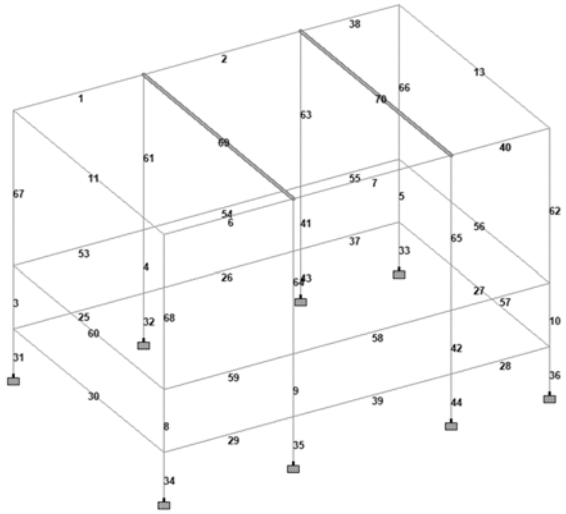


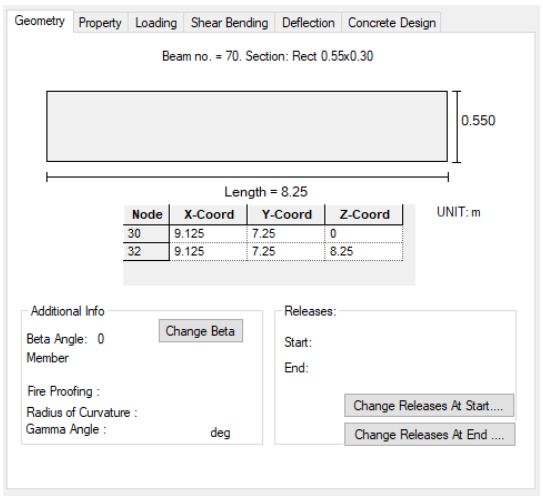
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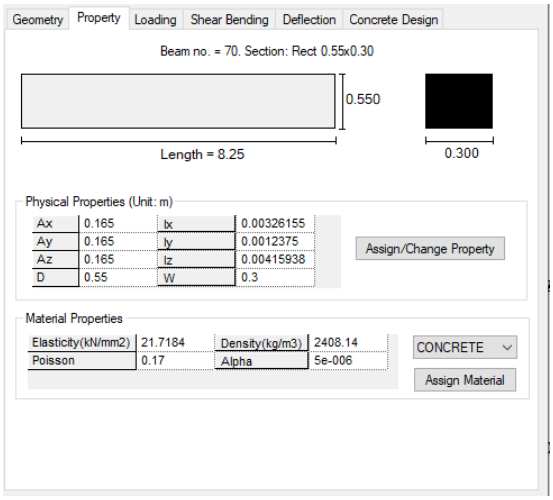
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9.3. Design of Beam(B2)







Hence Provide 3-20dia-Full Straight Bars Top & Bottom considering minimum reinfrocement requirements

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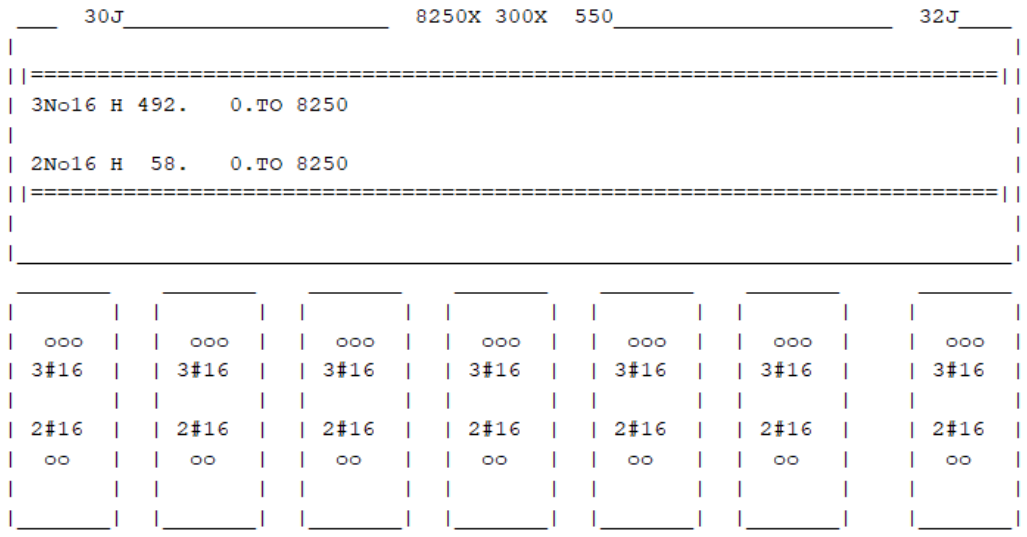
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9.3.1. Staad Output For B2

```
                B E A M  N O.    70 D E S I G N  R E S U L T S - SHEAR

   AT START SUPPORT - Vu=    14.16 KNS   Vc=  143.18 KNS   Vs=     0.00 KNS
   Tu=     0.00 KN-MET   Tc=     5.9 KN-MET   Ts=     0.0 KN-MET   LOAD      3
                     STIRRUPS ARE NOT REQUIRED.
   AT END   SUPPORT - Vu=    14.16 KNS   Vc=  143.18 KNS   Vs=     0.00 KNS
   Tu=     0.00 KN-MET   Tc=     5.9 KN-MET   Ts=     0.0 KN-MET   LOAD      3
                     STIRRUPS ARE NOT REQUIRED.
```



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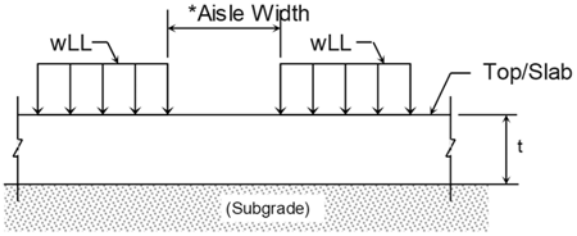
10. CONCRETE GRADE SLAB DESIGN

CONCRETE SLAB ON GRADE ANALYSIS

For Slab Subjected to Stationary Uniformly Distributed Live Loads

Input Data:

Slab Thickness, t =	5.91	in.	150.000	mm	
Concrete Strength, f 'c =	5000	psi	35.000	kpa	
Subgrade Modulus, k =	39.7866	pci	10800	Kn/m3	(AS PER SECTION-7.7-SOIL REPORT)
Factor of Safety, FS =	1.500		1.500		
Uniform Live Load, wLL =	208.90	psf	10.00	kpa	



Concrete Slab on Grade with Uniform Loads

Results:

Design Parameters:

Modulus of Rupture, MR =	636.40	psi	4387.79	kpa	MR = 9*SQRT(f 'c)
Allow. Bending Stress, Fb =	424.26	psi	2925.19	kpa	Fb = MR/FS
Modulus of Elasticity, Ec =	4030509	psi	27789329	kpa	Ec = 57000*SQRT(f 'c)
Poisson's Ratio, μ =	0.15		0.15		μ = 0.15 (assumed for concrete)
Radius of Stiffness, Lr =	36.52	in.	927.59	mm	Lr = (Ec*t^3/(12*(1-μ^2)*k))^0.25
Critical Aisle Width, Wcr =	6.72	ft.	2049.06	mm	Wcr = (2.209*Lr)/12

Stationary Uniformly Distributed Live Loads:

wLL(allow) =	835.34	psf	wLL(allow) = 257.876*Fb*SQRT(k*t/Ec)
	39.99	kpa	**wLL(allow) >= wLL, O.K.**

Provide Minimum Reinforcement of 0.14%@both faces = 0.14%(h) = 210 mm2/M

Table 24.4.3.2—Minimum ratios of deformed shrinkage and temperature reinforcement area to gross concrete area

Reinforcement type	f_y, psi	Minimum reinforcement ratio	
Deformed bars	< 60,000	0.0020	
Deformed bars or welded wire reinforcement	≥ 60,000	Greater of:	$\frac{0.0018 \times 60{,}000}{f_y}$
			0.0014

Provide WWF-8mm Dia@200 c-c Both Ways-T&B

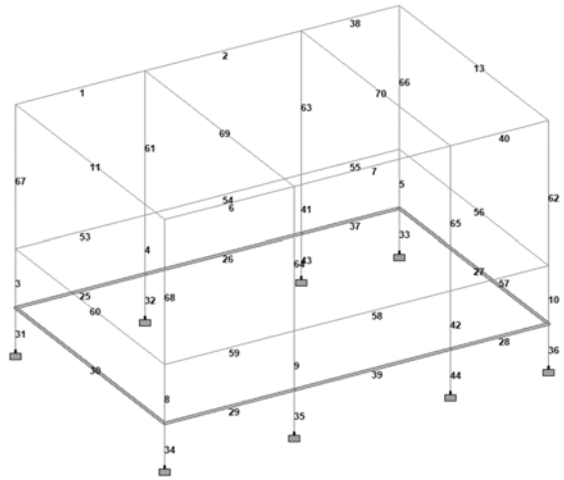
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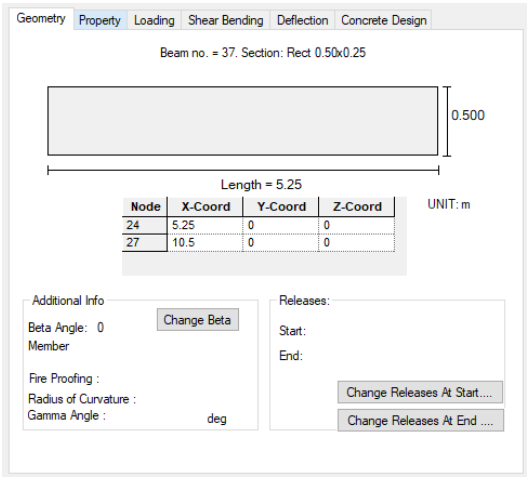
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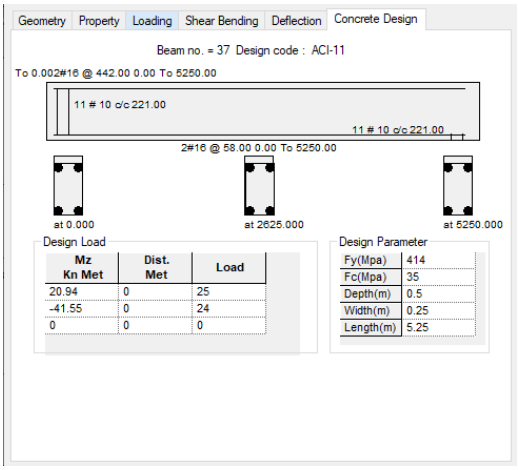
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11. DESIGN OF BEAM-(GB1)

11.1. DESIGN OF BEAM-(GB1)







Provide minimum 3-16 dia-Full Straight Bars Top & Bottom.

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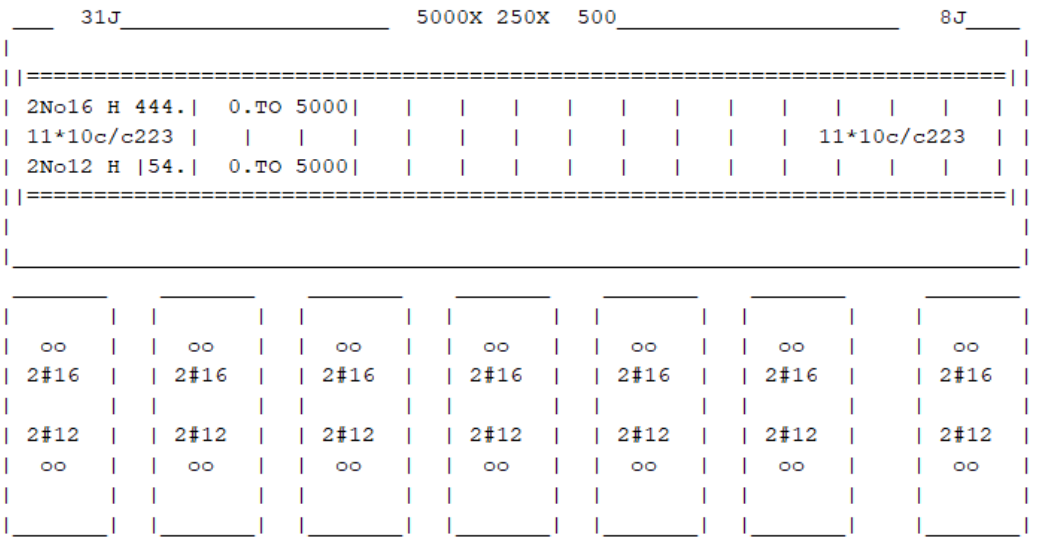
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11.1.1. STAAD OUTPUT FOR GB1

```
                B E A M   N O.    39 D E S I G N   R E S U L T S - SHEAR


 AT START SUPPORT - Vu=   43.01 KNS  Vc=  106.85 KNS  Vs=    0.00 KNS
 Tu=    0.02 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD    22
 NO STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 223. MM  C/C FOR 2058. MM

 AT END   SUPPORT - Vu=   43.15 KNS  Vc=  106.83 KNS  Vs=    0.00 KNS
 Tu=    0.02 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD    22
 NO STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 223. MM  C/C FOR 2058. MM
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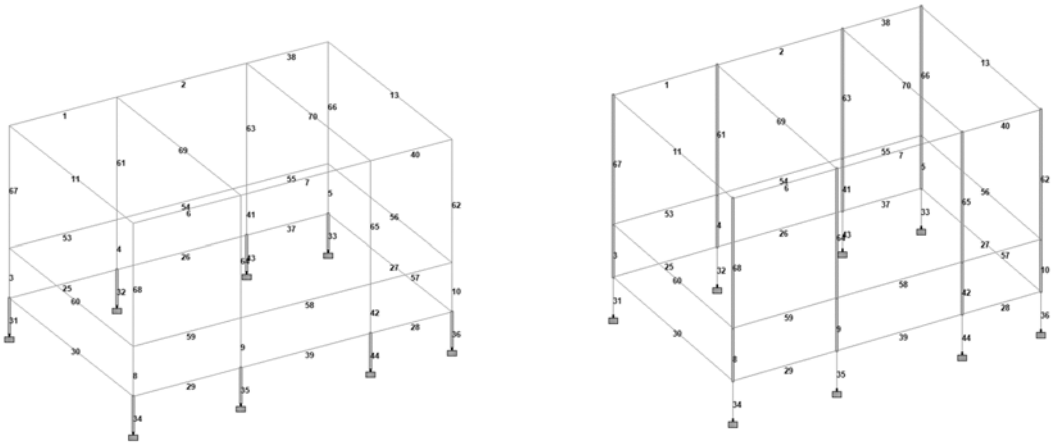


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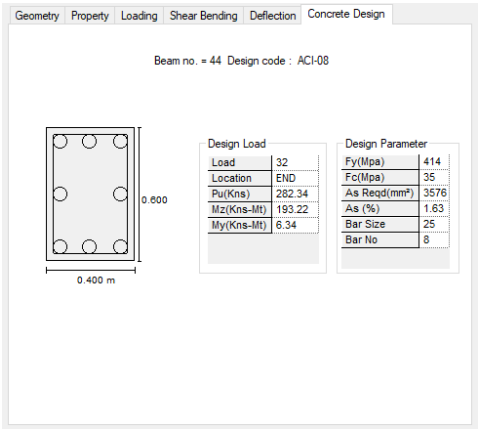
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12. CONCRETE COLUMN DESIGN

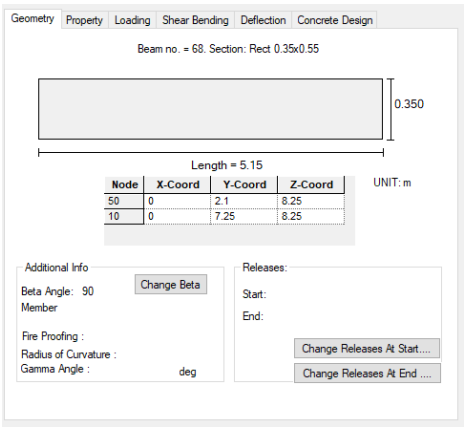


Column numbers

Column Design CR



Column Design C1



Hence Provide minimum 4-25 Dia@Corner+6-20 Dia@Middle for C1 & 10-25 Dia for Column Neck CR

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12.1. Staad Output For Column CR & C1

```
      COLUMN  NO.    44  DESIGN PER ACI 318-08 - AXIAL + BENDING


 FY - 413.7  FC -  35.0 MPA,  RECT SIZE - 600.0 X 400.0 MMS, TIED
        AREA OF STEEL REQUIRED =  3576.0  SQ. MM

 BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
 ------------------------------------------------------------

   8 - 25 MM               1.636       32      END     0.650
 (PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
 TIE BAR NUMBER   10 SPACING 400.00 MM



     COLUMN  NO.    68  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 413.7  FC -  35.0 MPA,  RECT SIZE - 550.0 X 350.0 MMS, TIED
        AREA OF STEEL REQUIRED =  3137.7  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
------------------------------------------------------------

  4 - 32 MM               1.671       26      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 350.00 MM


**********END OF COLUMN DESIGN RESULTS**********
```


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13. CANTILEVER SLAB DESIGN

DESIGN OF CANTILEVER SLAB

Geometrical Properties			Material Properties		
Span to center of support "sp_1"	0.8	met	Weight of reinforced concrete "γ_r	24	kN/m^3
Clear Span	0.6	met	Strength of concrete "f'_c"	35	N/mm^2
Clear span + depth of slab "sp_2"	0.75	met	Strength of steel "f_y"	420	N/mm^2
Max. eff span le = min (sp_1;sp_2)	0.8	met	Clear cover to concrete("c")	20	mm
Width of Slab "b"	1000	mm	Flexure strength reduction factor	0.9	
Thickness of slab "t"	150	mm	Shear strength reduction factor	0.75	
Eff. Depth "d"	125		Modulus of Elasticity of steel "Es"	2E+05	N/mm^2
Loading			Modulus of Elasticity of concrete	27806	N/mm^2
Self Weight of slab	3.6	kN/m^2	Modular Ratio "n = Es/Ec"	7.19	
Floor Finish	3.5	kN/m^2	**Provided Reinforcement**		
Wall load	0	kN/m^2	Top layer Nos.	6 Φ 10	mm
Total dead Load "U_{dl}"	7.1	kN/m^2	Bottom layer Nos.	0 Φ 12	mm
Live load	1	kN/m^2	Area of Top steel "$A_{s\ t}$"	468	mm^2
Total Ultimate Load "$Uu_{,tot}$"$_{(1.2D+1.6L)}$	10.12	kN/m^2	Area of Bottom steel "$A_{s\ b}$"	0	mm^2
Moment due to Overhang Point Load(0.125*0.15*24*0.6)	0.27	kNm	"d_b' = 10 mm		
Calculations					
Shear force Vu = U*le	**8.096**	**kN**			
Neg. Moment Mu = U*le^2/2	**3.51**	**kNm**			

Check for Section capacity								
Depth of comp. block "a" = $A_{s\ t}$*f_y/(0.85*f'_c*b)	6.61	mm						
Section capacity ΦMn = Φ*$A_{s\ t}$*f_y*(d-a/2)	**21.94**	**kNm**	**O.K**	>	**3.51**	**kNm**	(Mu)	
Shear requirements								
Shear capacity ΦVc = Φ*0.17*√f'c*b*d	**94.29**	**kN**	**O.K**	>	**8.096**	**kN**	(Vu)	

Serviceability requirements		
d' = c+d_b/2	26	mm
B = b/(n*$A_{s\ t}$)	0.297	
r = (n-1)*$A_{s\ b}$/(n*$A_{s\ t}$)	0	
k_d = (√(2*d*B(1+r*d'/d)+(1+r)^2)-(1+r))/B	29.01	mm
I_{cr} = bkd^3/3+n*$A_{s\ t}$*(d-k_d)^2+(n-1)*$A_{s\ b}$*(k_d-d'^{2}	3.915E+07	mm^4
Calculate Gross moment of inertia		
Area of slab A = t*b	2E+05	mm^2
Depth of neutral axis "y"	75	mm
Distance of top fiber from N.A "y_t"	75	mm
Gross moment of inertia "I_g"	2.813E+08	mm^4
Allowable stress fr = 0.62√f'c	3.67	mm^2
Cracking moment M_{cr} = fr*I_g/y_t	13.75	kNm

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Service moments

Moment due to dead load "M_{dl}"	2.54	kNm
Moment due to live load "Mll"	0.32	kNm
Total Moment Ms = $M_{dl}+M_{ll}$	2.86	kNm

Deflection due to dead + live loads

R(dl+ll) = M_{cr}/Ms	1.000		
I_e(dl+ll) = $R^3*I_g+(1-R^3)*I_{cr}$	2.813E+08	mm^4	
δ = K*5/48*Ms*le^2/(Ec*I_e)	0.08	mm	**K = 3.2 (conservative)**

Deflection due to dead load

R(dl) = M_{cr}/Md	5.411	
I_e(dl) = $R^3*I_g+(1-R^3)*I_{cr}$	3.839E+10	mm^4
δ = K*5/48*Md*le^2/(Ec*I_e)	0.00	mm

Deflection due to live load

Live load deflection = dl+ll deflection - dl deflection 0.08 mm

Allowable immediate deflection due to Live load =
per ACI 318 Tab 9.5b I_e/360 **2.22 mm > 0.08 mm O.K**

Long term deflection

% of liveload sustained "SL"	50	%
M_{sus} = $M_{dl}+SL*M_{ll}$	2.70	kNm
R(sus) = M_{cr}/Msus	5.091	
I_e(sus) = $R^3*I_g+(1-R^3)*I_{cr}$	3.198E+10	
δ = K*5/48*Msus*le^2/(Ec*I_e)	0.00	mm

Magnification factor for creep and shrinkage

$\lambda_\Delta = \xi / (1+50*\rho')$ 2.000

Where:-

ξ = 2 for 5 years or more cl.9.5.2.5 ACI 318

$\rho' = A_{s\ b}/(b*d)$ 0

Total deflection = λ_Δ*sus deflection + ll deflection 0.079 mm

Allowable deflection = I_e/240 3.333 mm > 0.079 mm O.K

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14. FOUNDATION DESIGN

Foundations have been designed based on the Soil Report Recommendations. The minimum Embedment Depth of Foundations has been considered as 1.2m & applicable SBC of 350 Kpa. (Refer Doc-MD-216-3000-EG-CV-RPT-1008 for Details)

TABLE 4: RECOMMENDED NET ALLOWABLE BEARING CAPACITIES

PRPJECT: MANSOURAH - MANSSARAH GOLD PROJECT
CLIENT: L & T CONSTRUCTION

Area	Borehole No.	Proposed Structures	Foundation Type	Depth of Foundation below FGL	Footing Size (B x L)	Net Allowable Bearing Capacity Based on		Recommended Net Allowable Bearing Capacity	Estimated Total Settlement	Vertical Modulus of Subgrade Reaction (Ks)	Horizontal Modulus of Subgrade Reaction (Kh)
						Settlement	Shear Failure				
				(m)	(m)	(kN/m^2)	(kN/m^2)	(kN/m^2)	(mm)	($kN/m^2/m$)	(kN/m^2)
Village Area	BH-21	Village	Isolated Square	1.20	1.0 x 1.0	1000	350	350	25	42,000	32,200
				1.50		1100	500	500	25	60,000	38,900
				1.20	2.0 x 2.0	700	580	580	25	69,600	37,500
				1.50		850	750	750	25	90,000	44,200
				1.20	3.0 x 3.0	600	750	600	25	72,000	42,800
				1.50		700	930	700	25	84,000	49,500
				1.20	4.0 x 4.0	550	910	550	25	66,000	48,100
				1.50		625	1090	625	25	75,000	54,800
				1.20	5.0 x 5.0	475	1060	475	25	57,000	53,400
				1.50		550	1240	550	25	66,000	60,200

Refer Attachment- II for Foundation Design calculations

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REFERENCES

1.)	BUILDING PLANS	MD-216-1800-EG-AR-GAD-3152
2.)	OVERALL LAYOUT PLAN	MD-216-1800-EG-CV-GAD-1565
3.)	LAYOUT & DETAILS OF FOUNDATION, COLUMN & GRADE BEAM	MD-216-1800-EG-CV-GAD-1546
4.)	LAYOUT & DETAILS OF ROOF & GRADE SLAB	MD-216-1800-EG-CV-GAD-1569
5.)	GEOTECHNICAL REPORT	MD-216-3000-EG-CV-RPT-1008
6.)	STRUCTURAL DESIGN CRITERIA	MD-216-0000-EG-ST-CRT-0001

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ATTACHMENT 1 STAAD INPUT DATA

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STAAD SPACE
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 18-Oct-19
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 0 0; 2 4.125 0 0; 3 12.25 0 0; 4 0 7.25 0; 5 4.125 7.25 0; 6 12.25 7.25 0;
7 0 0 8.25; 8 4.125 0 8.25; 9 12.25 0 8.25; 10 0 7.25 8.25; 11 4.125 7.25 8.25;
12 12.25 7.25 8.25; 23 0 -1.5 0; 24 4.125 -1.5 0; 25 12.25 -1.5 0;
26 0 -1.5 8.25; 27 4.125 -1.5 8.25; 28 12.25 -1.5 8.25; 29 9.125 0 0;
30 9.125 7.25 0; 31 9.125 0 8.25; 32 9.125 7.25 8.25; 33 9.125 -1.5 0;
34 9.125 -1.5 8.25; 43 4.125 2.1 0; 44 12.25 2.1 8.25; 45 9.125 2.1 0;
46 4.125 2.1 8.25; 47 9.125 2.1 8.25; 48 12.25 2.1 0; 49 0 2.1 0;
50 0 2.1 8.25;
MEMBER INCIDENCES
1 4 5; 2 5 30; 3 1 49; 4 2 43; 5 3 48; 6 10 11; 7 11 32; 8 7 50; 9 8 46;
10 9 44; 11 4 10; 13 6 12; 25 1 2; 26 2 29; 27 3 9; 28 9 31; 29 8 7; 30 7 1;
31 1 23; 32 2 24; 33 3 25; 34 7 26; 35 8 27; 36 9 28; 37 29 3; 38 30 6;
39 31 8; 40 32 12; 41 29 45; 42 31 47; 43 29 33; 44 31 34; 53 49 43; 54 43 45;
55 45 48; 56 48 44; 57 44 47; 58 47 46; 59 46 50; 60 50 49; 61 43 5; 62 44 12;
63 45 30; 64 46 11; 65 47 32; 66 48 6; 67 49 4; 68 50 10; 69 5 11; 70 30 32;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17184e+007
POISSON 0.17
DENSITY 23.6158
ALPHA 5e-006
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27578.9
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
3 TO 5 8 TO 10 41 42 61 TO 68 PRIS YD 0.35 ZD 0.55
1 2 6 7 38 40 PRIS YD 0.5 ZD 0.25
MEMBER PROPERTY AMERICAN
25 TO 30 37 39 PRIS YD 0.5 ZD 0.25
MEMBER PROPERTY AMERICAN
11 13 PRIS YD 1 ZD 0.3
MEMBER PROPERTY AMERICAN
53 TO 60 PRIS YD 0.25 ZD 0.25
31 TO 36 43 44 PRIS YD 0.4 ZD 0.6
MEMBER PROPERTY AMERICAN
69 70 PRIS YD 0.55 ZD 0.3
CONSTANTS
BETA 90 MEMB 3 TO 5 8 TO 10 31 TO 36 41 TO 44 61 TO 68
MATERIAL CONCRETE ALL
SUPPORTS
23 TO 28 33 34 FIXED
****
MEMBER CRACKED
3 TO 5 8 TO 10 31 TO 36 41 TO 44 61 TO 67 -
68 REDUCTION RIX 0.700000 RIY 0.700000 RIZ 0.700000
1 2 6 7 11 13 25 TO 30 37 TO 40 53 TO 60 69 -
70 REDUCTION RIX 0.350000 RIY 0.350000 RIZ 0.350000
DEFINE IBC 2003
SDS 0.106 SD1 0.032 S1 0.02 IE 1 RX 2.5 RZ 2.5 SCLASS 2
SELFWEIGHT 1

```
MEMBER WEIGHT
1 2 6 7 38 40 UNI 26
DEFINE WIND LOAD
TYPE 1 WIND X WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 8.000 M 30.000 M 30.000 M 2.000 0.010 0 -
0 0 0 0 1.136 1.000 1.000 0.850 0 -
0 0 0 0.868 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.68826 1.68826 1.69709 1.70553 1.71362 1.72139 1.72886 1.73606 1.74301 -
 1.74972 1.75623 1.76253 1.76864 1.77458 1.78035 HEIG  0  4.572  4.83569 -
 5.09938  5.36308  5.62677  5.89046  6.15415  6.41785  6.68154  6.94523 -
 7.20892  7.47261  7.73631  8
TYPE 2 WIND X LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 8.000 M 30.000 M 30.000 M 2.000 0.010 1 -
0 0 0 0 1.136 1.000 1.000 0.850 0 -
0 0 0 0.868 -0.500 0.550
!> END GENERATED DATA BLOCK
INT -1.40788 -1.40788 HEIG  0  8
TYPE 3 SIDEWARD X
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 8.000 M 30.000 M 30.000 M 2.000 0.010 2 -
0 0 0 0 1.136 1.000 1.000 0.850 0 -
0 0 0 0.868 -0.700 0.550
!> END GENERATED DATA BLOCK
INT -1.65619 -1.65619 HEIG  0  8
TYPE 4 WIND Z WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 8.000 M 30.000 M 30.000 M 2.000 0.010 0 -
0 0 0 0 1.136 1.000 1.000 0.850 0 -
0 0 0 0.868 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.68826 1.68826 1.69709 1.70553 1.71362 1.72139 1.72886 1.73606 1.74301 -
 1.74972 1.75623 1.76253 1.76864 1.77458 1.78035 HEIG  0  4.572  4.83569 -
 5.09938  5.36308  5.62677  5.89046  6.15415  6.41785  6.68154  6.94523 -
 7.20892  7.47261  7.73631  8
TYPE 5 WIND Z LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 8.000 M 30.000 M 30.000 M 2.000 0.010 1 -
0 0 0 0 1.136 1.000 1.000 0.850 0 -
0 0 0 0.868 -0.500 0.550
!> END GENERATED DATA BLOCK
INT -1.40788 -1.40788 HEIG  0  8
TYPE 6 SIDEWARD Z
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 8.000 M 30.000 M 30.000 M 2.000 0.010 2 -
0 0 0 0 1.136 1.000 1.000 0.850 0 -
0 0 0 0.868 -0.700 0.550
!> END GENERATED DATA BLOCK
INT -1.65619 -1.65619 HEIG  0  8
LOAD 1 LOADTYPE Seismic  TITLE EQK LOAD Z
```

```
IBC LOAD Z 1
PERFORM ANALYSIS
CHANGE
LOAD 2 LOADTYPE Seismic  TITLE EQK LOAD X
IBC LOAD X 1
PERFORM ANALYSIS
CHANGE
LOAD 3 LOADTYPE Dead  TITLE DL
SELFWEIGHT Y -1
MEMBER LOAD
25 TO 30 37 39 53 TO 60 UNI GY -12
MEMBER LOAD
1 2 6 7 38 40 UNI GY -26
11 UMOM GZ 1
13 UMOM GZ -1
1 2 38 UMOM GX -1
6 7 40 UMOM GX 1
LOAD 4 LOADTYPE Live  TITLE LL
MEMBER LOAD
1 2 6 7 38 40 UNI GY -3
11 UMOM GZ 0.2
13 UMOM GZ -0.2
1 2 38 UMOM GX -0.2
6 7 40 UMOM GX 0.2
11 13 69 70 CON GY -7.5 5 0.15
11 CMOM GZ 0.15 5 0.15
13 CMOM GZ -0.15 5 0.15
LOAD 5 LOADTYPE Wind  TITLE WL+X
WIND LOAD X 1 TYPE 1 YR 0 8
WIND LOAD -X 1 TYPE 2 YR 0 8
WIND LOAD -Z 1 TYPE 3 YR 0 8
WIND LOAD -Z -1 TYPE 3 YR 0 8
LOAD 6 LOADTYPE Wind  TITLE WL+Z
WIND LOAD Z 1 TYPE 4 YR 0 8
WIND LOAD -Z 1 TYPE 5 YR 0 8
WIND LOAD -X 1 TYPE 6 YR 0 8
WIND LOAD -X -1 TYPE 6 YR 0 8
LOAD 7 LOADTYPE Temperature  TITLE TEMP
TEMPERATURE LOAD
1 TO 11 13 38 40 TO 42 53 TO 70 TEMP 25
LOAD COMB 8 SLS: (DL)
3 1.0
LOAD COMB 9 SLS: (DL+T)
3 1.0 7 1.0
LOAD COMB 10 SLS: (D+LR)
3 1.0 4 1.0
LOAD COMB 11 SLS: (D+0.75LR)
3 1.0 4 0.75
LOAD COMB 12 SLS: (D+0.7EZ)
3 1.0 1 0.7
LOAD COMB 13 SLS: (D+0.7EX)
3 1.0 2 0.7
LOAD COMB 14 SLS: (D+0.75WX+0.75LR)
3 1.0 5 0.75 4 0.75
LOAD COMB 15 SLS: (D+0.75WZ+0.75LR)
3 1.0 6 0.75 4 0.75
LOAD COMB 16 SLS: (0.6D+WX)
3 0.6 5 1.0
```

LOAD COMB 17 SLS: (0.6D+WZ)
3 0.6 6 1.0
LOAD COMB 18 SLS: (0.6D+0.7EX)
3 0.6 2 0.7
LOAD COMB 19 SLS: (0.6D+0.7EZ)
3 0.6 1 0.7
LOAD COMB 20 SLS: (D+WX)
3 1.0 5 1.0
LOAD COMB 21 SLS: (D+WZ)
3 1.0 6 1.0
LOAD COMB 22 ULS: (1.4 DL)
3 1.4
LOAD COMB 23 ULS: (1.4D+1.4T+0.5LR)
3 1.4 4 0.5 7 1.4
LOAD COMB 24 ULS: (1.4D+1.4T+0.5LR)
3 1.4 4 0.5 7 1.4
LOAD COMB 25 ULS: (1.2D+0.80WX+1.6LR)
3 1.2 5 0.8 4 1.6
LOAD COMB 26 ULS: (1.2D+1.6WZ+0.5LR)
3 1.2 6 1.6 4 0.5
LOAD COMB 27 ULS: (1.2D+1.6WX+0.5LR)
3 1.2 4 0.5 5 1.6
LOAD COMB 28 ULS: (1.2D+1.6WZ+0.5LR)
3 1.2 4 0.5 6 1.6
LOAD COMB 29 ULS: (1.2D+EX)
3 1.2 2 1.0
LOAD COMB 30 ULS: (1.2D+EZ)
3 1.2 1 1.0
LOAD COMB 31 ULS: (0.90D+1.6WX)
3 0.9 5 1.6
LOAD COMB 32 ULS: (0.90D+1.6WZ)
3 0.9 6 1.6
LOAD COMB 33 ULS: (0.90D+EZ)
3 0.9 1 1.0
LOAD COMB 34 ULS: (0.90D+EX)
3 0.9 2 1.0
PERFORM ANALYSIS
START CONCRETE DESIGN
CODE ACI
FC 35000 ALL
EFACE 0.44 MEMB 1 2 6 7 38 40
SFACE 0.44 MEMB 1 2 6 7 38 40
EFACE 0.29 MEMB 11 13
SFACE 0.29 MEMB 11 13
CLB 0.04 MEMB 1 TO 11 13 27 30 TO 36 38 40 TO 44 53 TO 70
CLS 0.04 ALL
CLT 0.04 MEMB 1 TO 11 13 27 30 TO 36 38 40 TO 44 53 TO 70
MINSEC 10 ALL
MINMAIN 16 MEMB 11 13 69 70
MINMAIN 25 MEMB 3 TO 5 8 TO 10 31 TO 36 41 TO 44
DESIGN BEAM 1 2 6 7 11 13 25 TO 30 37 TO 40 69 70
DESIGN COLUMN 3 TO 5 8 TO 10 31 TO 36 41 TO 44 61 TO 68
END CONCRETE DESIGN
FINISH

ATTACHMENT 2 FOUNDATION DESIGN CALCULATIONS

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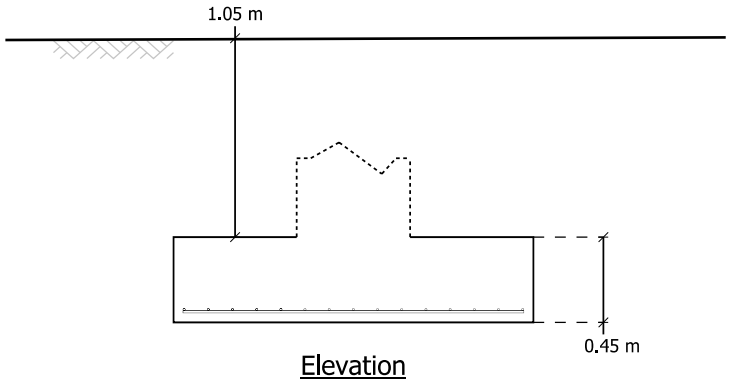
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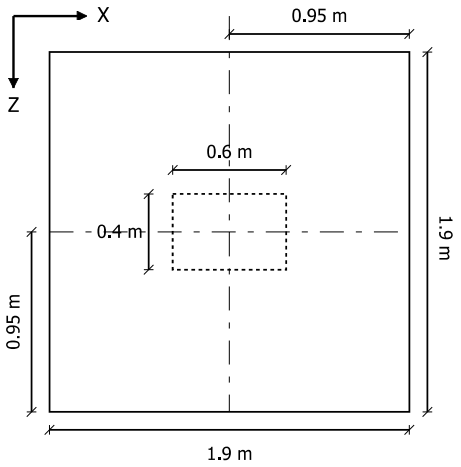
Isolated Footing Design(ACI 318-11)

Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
24	1	1.900m	1.900m	0.450m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
24	14 - ϕ12	15 - ϕ12	N/A	N/A	N/A	N/A



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 450.000mm
Footing Length - X (Fl) : 1900.000mm
Footing Width - Z (Fw) : 1900.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.600m

Column Width - Z (B_{col}) : 0.400m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 415.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #20
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #20
Pedestal Minimum Bar Size : #12
Pedestal Maximum Bar Size : #22
Minimum Bar Spacing : 50.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 1050.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.500
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.900m

Initial Width (W_o) = 1.900m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
8	SLS: (DL)	1.00	1.00	1.00
9	SLS: (DL+T)	1.00	1.00	1.00
10	SLS: (D+LR)	1.00	1.00	1.00
11	SLS: (D+0.75LR)	1.00	1.00	1.00
12	SLS: (D+0.7EZ)	1.00	1.00	1.00
13	SLS: (D+0.7EX)	1.00	1.00	1.00
14	SLS: (D+0.75WX+0.75LR)	1.00	1.00	1.00
15	SLS: (D+0.75WZ+0.75LR)	1.00	1.00	1.00
16	SLS: (0.6D+WX)	1.00	1.00	1.00
17	SLS: (0.6D+WZ)	1.00	1.00	1.00
18	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
19	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
20	SLS: (D+WX)	1.00	1.00	1.00
21	SLS: (D+WZ)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
22	ULS: (1.4 DL)	1.00	1.00	1.00
23	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
24	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
25	ULS: (1.2D+0.80WX+1.6LR)	1.00	1.00	1.00
26	ULS: (1.2D+1.6WZ+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+1.6WX+0.5LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+EX)	1.00	1.00	1.00
30	ULS: (1.2D+EZ)	1.00	1.00	1.00
31	ULS: (0.90D+1.6WX)	1.00	1.00	1.00
32	ULS: (0.90D+1.6WZ)	1.00	1.00	1.00
33	ULS: (0.90D+EZ)	1.00	1.00	1.00
34	ULS: (0.90D+EX)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
8	325.150	-4.253	-1.894	-5.573	1.183
9	324.085	-2.545	-2.133	-6.842	3.545
10	342.159	-4.250	-2.530	-7.761	1.189
11	337.907	-4.251	-2.371	-7.214	1.188
12	323.116	-4.282	0.726	7.320	1.265
13	326.934	-0.962	-1.894	-5.573	-8.266
14	343.755	15.222	-22.610	-47.570	-42.963
15	329.001	-4.763	30.749	99.159	2.502
16	202.887	23.411	-28.122	-57.152	-58.158
17	183.215	-3.235	43.024	138.487	2.462
18	196.874	0.739	-1.136	-3.344	-8.739
19	193.056	-2.581	1.483	9.549	0.792
20	332.947	21.710	-28.879	-59.381	-57.685
21	313.275	-4.936	42.267	136.258	2.935

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
22	455.209	-5.954	-2.651	-7.803	1.656
23	462.224	-3.561	-3.305	-10.672	4.966
24	462.224	-3.561	-3.305	-10.672	4.966

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
25	423.632	15.672	-24.879	-53.234	-45.664
26	379.685	-6.195	68.066	219.148	4.227
27	411.160	36.439	-45.768	-93.874	-92.765
28	379.685	-6.195	68.066	219.148	4.227
29	392.729	-0.402	-2.272	-6.688	-12.078
30	387.274	-5.145	1.469	11.731	1.536
31	305.110	37.713	-44.882	-91.109	-93.123
32	273.635	-4.920	68.953	221.914	3.869
33	289.729	-3.869	2.037	13.403	1.182
34	295.184	0.873	-1.704	-5.016	-12.433

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 3.610 m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 2.447 m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.900 m Governing Load Case : # 8

Width (W_2) = 1.900 m Governing Load Case : # 8

Depth (D_2) = 0.450 m Governing Load Case : # 27

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 3.610 m^2

Final Soil Height = 1.050 m

Footing Self Weight = 38.987 kN

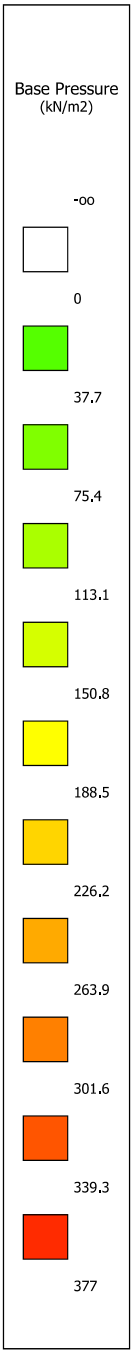
Gross Soil Bearing Capacity = 377.00 kN/m2

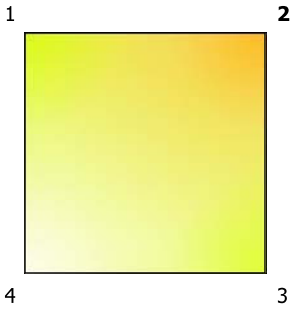
Soil Weight On Top Of Footing = 63.691 kN

Pressures at 4 Corners

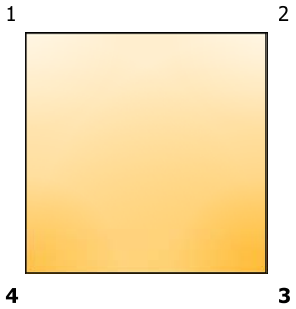
Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner

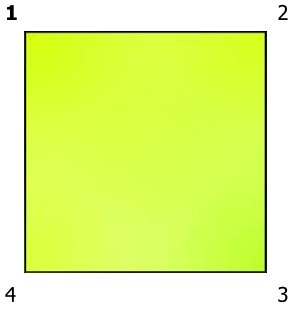




Corner 2

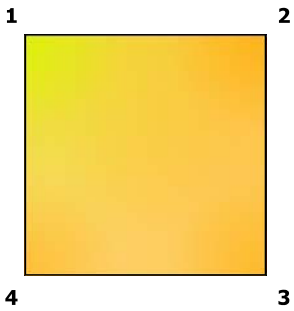


Corner 3, 4



Corner 1

Governing Base Pressure Contour



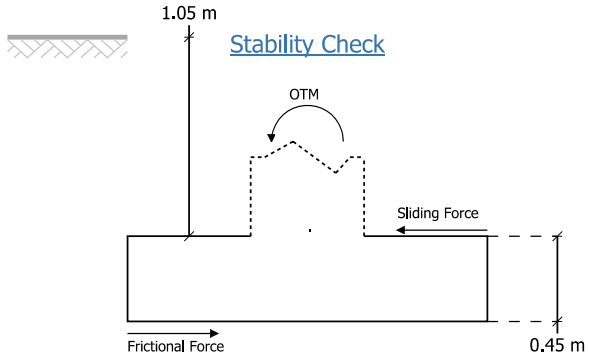
Corner 1, 2, 3, 4

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
10	**133.7213**	128.2949	112.7254	118.1519	0.000
20	124.9780	**242.9924**	116.3651	-1.6494	0.000
21	-16.1005	-25.1219	**246.5450**	255.5664	0.300
21	-16.1005	-25.1219	246.5450	**255.5664**	0.300

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
10	**133.7213**	128.2949	112.7254	118.1519
20	124.9779	**242.9926**	116.3650	0.0000
21	0.0000	0.0000	**248.7381**	258.6239
21	0.0000	0.0000	248.7381	**258.6239**



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
8	50.298	112.968	64.982	1.500	63.700	131.609	1.500
9	83.859	100.032	90.884	1.500	52.334	86.677	1.500
10	52.336	87.913	63.597	1.500	47.809	136.594	1.500
11	51.826	92.916	64.009	1.500	50.892	135.350	1.500
12	49.718	293.444	69.324	1.500	53.276	127.074	1.500
13	223.247	113.439	143.022	1.500	63.964	52.243	1.500
14	14.664	9.872	11.582	1.500	7.394	8.536	1.500
15	45.317	7.019	9.810	1.500	3.655	88.511	1.500
16	6.526	5.433	5.905	1.500	4.200	4.241	1.500
17	44.191	3.322	4.685	1.500	1.739	69.597	1.500
18	202.681	131.828	156.283	1.500	74.558	31.489	1.500
19	57.293	99.712	70.253	1.500	27.780	144.396	1.500
20	10.033	7.542	8.526	1.500	5.757	6.151	1.500
21	42.136	4.921	6.912	1.500	2.563	76.842	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 16
Governing Disturbing Force : 23.411kN
Governing Restoring Force : 152.782kN
Minimum Sliding Ratio for the Critical Load Case : 6.526
Critical Load Case for Overturning about X-Direction : 17
Governing Overturning Moment : 157.848kNm
Governing Resisting Moment : 274.466kNm
Minimum Overturning Ratio for the Critical Load Case : 1.739

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 17

Governing Disturbing Force : 43.024kN

Governing Restoring Force : 142.946kN

Minimum Sliding Ratio for the Critical Load Case : 3.322

Critical Load Case for Overturning about Z-Direction : 16

Governing Overturning Moment : -68.692kNm

Governing Resisting Moment : 291.358kNm

Minimum Overturning Ratio for the Critical Load Case : 4.241

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 17

Governing Disturbing Force : 43.146kN

Governing Restoring Force : 202.157kN

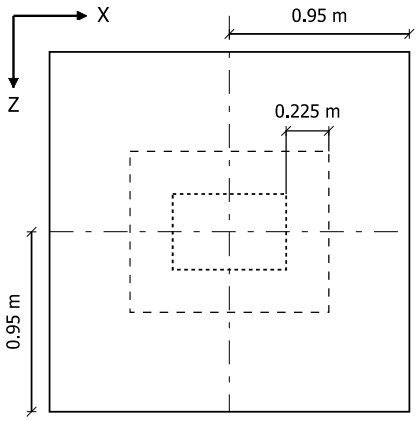
Minimum Sliding Ratio for the Critical Load Case : 4.685

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.450m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.394m

For rectangular column, β_c = B_{col} / D_{col} = 1.500

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 668.166kN, Load Case # 32

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 3.576m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 3229.934kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 4434.582kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2768.515kN

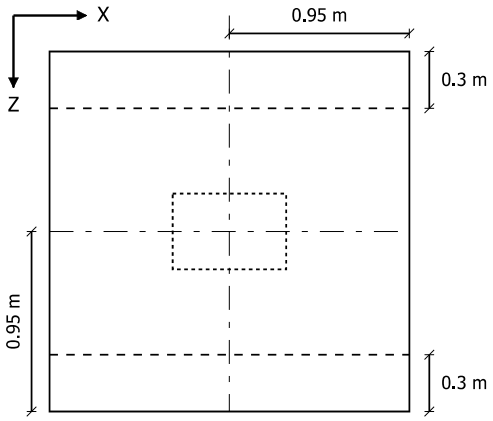
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 2076.386kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 735.484kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{zd}$ = 0.356m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

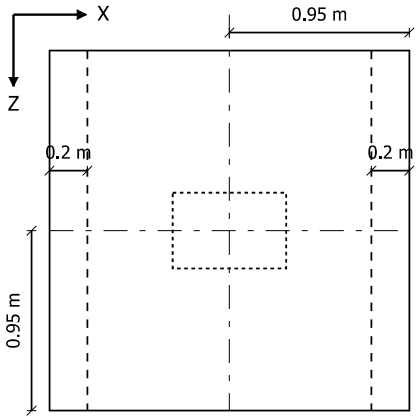
From above calculations, 0.75 X V_c = 551.613 kN

Critical load case for V_{ux} is # 26 $V_{ux} = V_{ux}|_{x=D_x}$ = 180.722 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 735.484 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.256 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

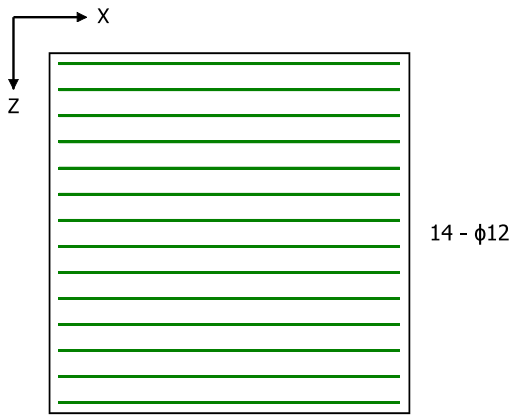
From above calculations, 0.75 X V_c = 551.613 kN

Critical load case for V_{uz} is # 27 $V_{uz} = V_{uz}|_{z=D_z}$ = 107.066 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 27

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.394 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03374
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02530
From ACI Cl. 7.12.2, ρ_{min} =		0.00185
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	13.950

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	1.250 m
Ultimate moment,	$M_u\big\vert_{z=D_z}$ =	89.405 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	99.338 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00082
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00071
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1582.856 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 50.000mm

Selected spacing (S) = 137.538mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

According to ACI 318 Clause No- 10.6.4 1st

Max spacing for Cracking Consideration = 254.793mm

Safe for Cracking Aspect.

Based on spacing reinforcement increment; provided reinforcement is

]

#12 @ 135mm o.c.

Required development length for bars =	$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ =	=0.406	m
Available development length for bars, D_L =	$0.5 \times (L - D_{col}) \cdot - C_{cover}$ =	0.600	m
Try bar size # 12	Area of one bar =	113.097	mm2
Number of bars required, N_{bar} =	$\frac{A_s}{A_{bar}}$ =	14	

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	1583.352	mm2
d_{eff} =	D - C_{cover} - 0.5 X (dia. of one bar) =	0.394	m
Reinforcement ratio, ρ =	$\frac{A_{s_total}}{(d_{eff} \times W)}$ =	0.00212	

From ACI Cl.7.6.1, minimum req'd clear distance between bars

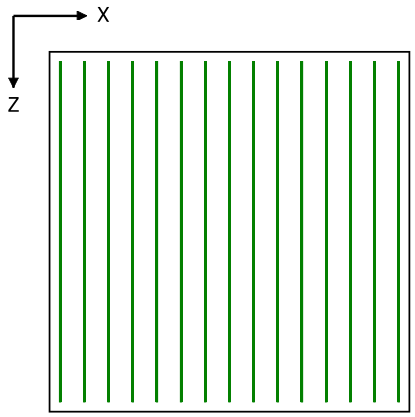
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 50.000mm

Provided Steel Area / Required Steel Area = 1.000

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



15 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 32

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.382	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03374	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02530	
From ACI Cl.7.12.2, ρ_{min} =		0.00189	
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{(0.85 \times F_c')}$ =	13.950	

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z =	$0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ =	1.150	m
Ultimate moment,	$M_u\|_{x=D_x}$ =	197.758	kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	219.731	kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00194	
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00164	
Since	ρ= ρ_{min}	ρ_{min} Governs	
Area of Steel Required, A_s =	ρ × W × d_{eff} =	1620.172	mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 50.000mm

Selected spacing (S) = 127.714mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

According to ACI 318 Clause No- 10.6.4 3rd

Max spacing for Cracking Consideration = 254.793mm

Safe for Cracking Aspect.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 125mm o.c.

Required development length for bars =	$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ =	=0.406	m
Available development length for bars, D_L =	$0.5 \times (L - D_{col}) \cdot - C_{cover}$ =	0.700	m
Try bar size # 12	Area of one bar =	113.097	mm2
Number of bars required, N_{bar} =	$\frac{A_s}{A_{bar}}$ =	15	

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	1696.448	mm2
d_{eff} =	D - C_{cover} - 1.5 X (dia. of one bar) =	0.382	m
Reinforcement ratio, ρ =	$\frac{A_{s_total}}{(d_{eff} \times W)}$ =	0.00234	

From ACI Cl.7.6.1, minimum req'd clear distance between bars

C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 50.000mm

Provided Steel Area / Required Steel Area = 1.047

Check to see if width is sufficient to accomodate bars

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Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	1.9 m	1.9 m	0.45 m	N/A	N/A	15 - ϕ12	14 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet